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1.4.1 Conditioning of a linear system

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saying, "Let's bring the norm of x to the left the norm of b to the right," And rewrite this as 1 over the norm of x is less than or equal to the norm of A times 1 over the norm of b. That puts us partially towards this right here. Alright. Now, we need to somehow relate delta x and delta b, the norms And for that we're actually going to rewrite A times delta x is equal to Delta b as delta x is equal to A inverse times delta b

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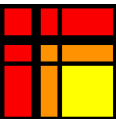
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Reading assignment

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Read Unit 1.4.1 of the notes. [\[LINK\]](#)

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Homework 1.4.1.1

1/1 point (graded)

Let $\|\cdot\|$ be a vector norm and corresponding induced matrix norm.

$\|I\| = 1.$

TRUE

▼

✔ Answer: TRUE

For full solution see notes: [[LINK](#)]

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Answers are displayed within the problem

Homework 1.4.1.2

1/1 point (graded)

Let $\|\cdot\|$ be a vector norm and corresponding induced matrix norm, and A a nonsingular matrix.

$\kappa(A) = \|A\|\|A^{-1}\| \geq 1.$

TRUE

▼

✔ Answer: TRUE

For full solution see notes: [[LINK](#)]

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